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What's New in C#

Jason Bock
Staff Software Engineer
Rocket Mortgage

Level: Intermediate

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Downloads

https://github.com/JasonBock/WhatsNewInCSharp

Overview

- Language Evolution
- Using Modern C# Features
- Call to Action

Remember…
https://github.com/JasonBock/WhatsNewInCSharp

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What’s New in C#

LANGUAGE EVOLUTION

C#

Est. 2002

Version Number	Release Date	Feature Count
1.0	1/2002	22
1.2	4/2003	2
2	11/2005	11
3	11/2007	10
4	4/2010	4
5	8/2012	3
6	6/2015	12
7.0	3/2017	12
7.1	8/2017	5
7.2	11/2017	6
7.3	5/2018	9
8.0	9/2019	14
9.0	11/2020	17
10.0	11/2021	16
11.0	11/2022	16
12.0	11/2023	7
13.0	11/2024	10
14.0	11/2024	11

Version Number	Release Date	Feature Count
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7.0	3/2017	12
7.1	8/2017	5
7.2	11/2017	6
7.3	5/2018	9
8.0	9/2019	14
9.0	11/2020	17
10.0	11/2021	16
11.0	11/2022	16
12.0	11/2023	7
13.0	11/2024	10
14.0	11/2024	11

The .NET Compiler Platform

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Roslyn is the open-source implementation of both the C# and Visual Basic compilers with an API surface for building code analysis tools.

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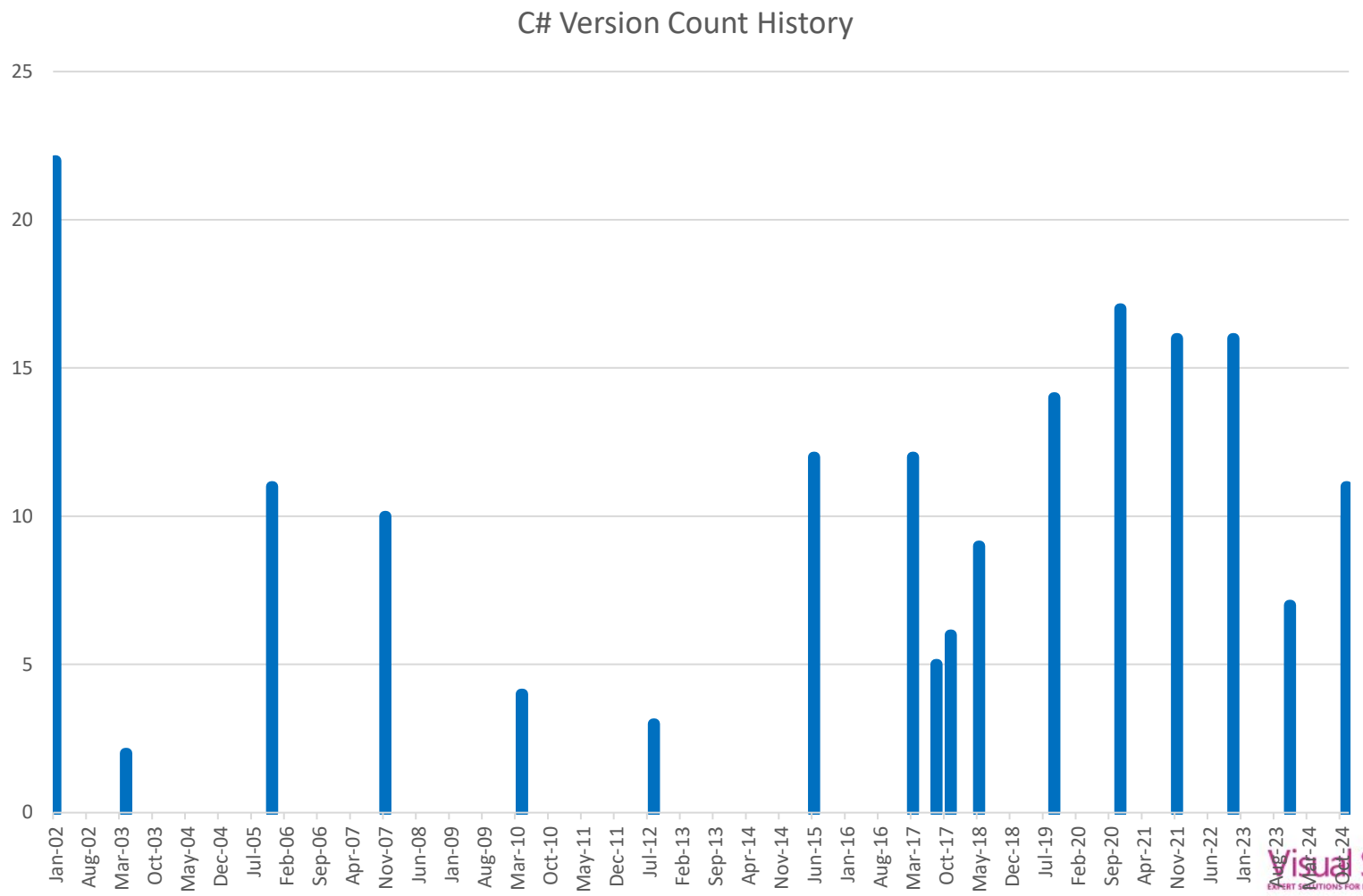

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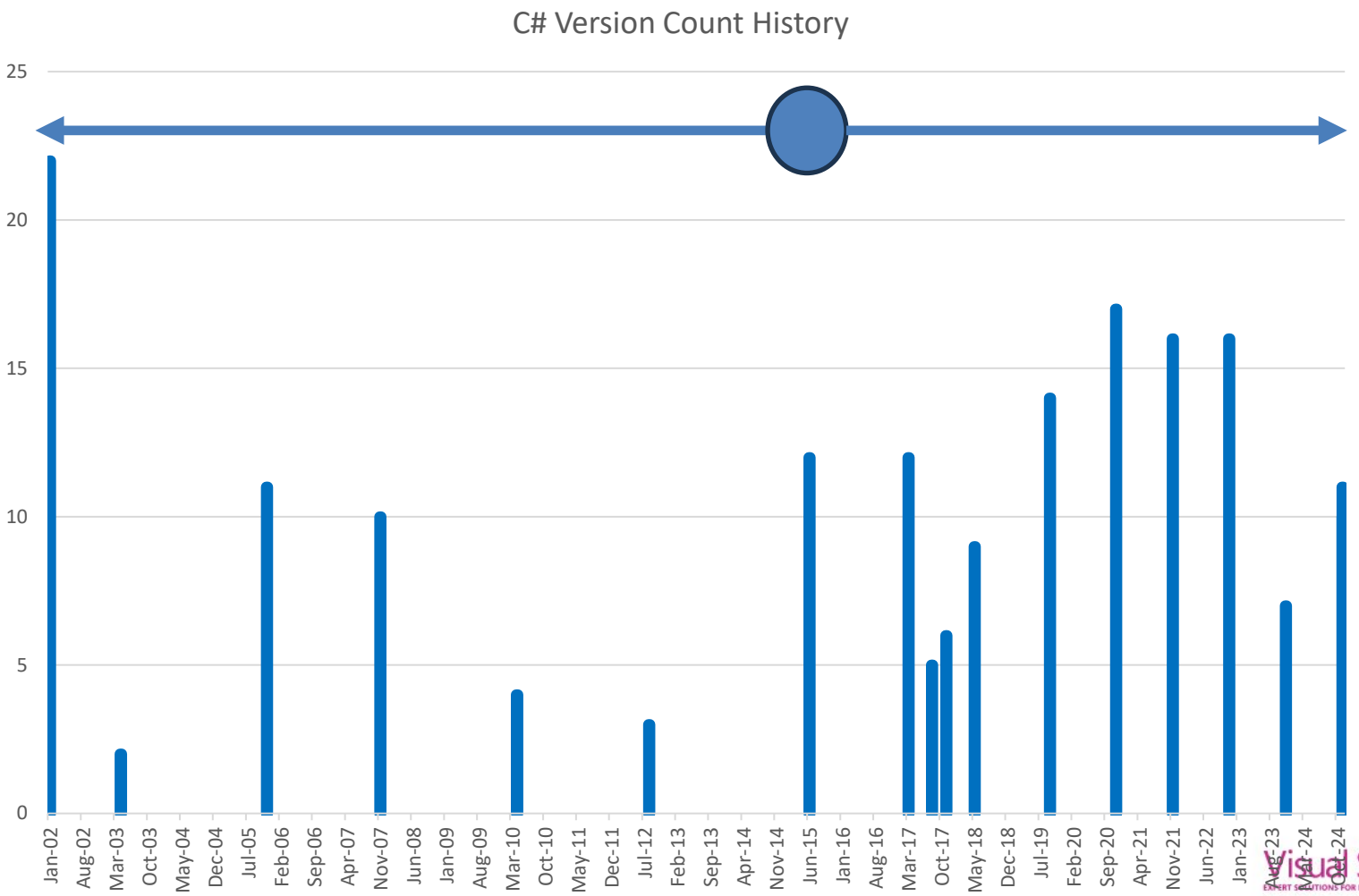

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Before Roslyn

52 features in 10.5 years

After Roslyn

135 features in 10.5 years

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Before Roslyn
5 features a year

After Roslyn
13 features a year

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DEMO: USING MODERN C# FEATURES

Features

.editorconfig

Directory.Build.props

Directory.Packages.props

.csproj (SDK style) and .slnx

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Features

- Required Properties
- Tuples
- Nullable Reference Types
- Patterns
- Raw String Literals
- Static Abstract Members in Interfaces
- Target-Type New
- Object Deconstruction
- Records
- Source Generators
- Collection Expressions
- Extensions
- field Keyword
- Unbound Generic Types in nameof()

What’s New in C#

CALL TO ACTION

C# 15

- Dictionary expressions
- Unions
- Closed class hierarchies
- Unsafe evolution
- Extension indexers
- ExtendedLayoutAttribute
- Runtime Async
- Collection expression arguments

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Numeric IntPtr

Function pointers

Unsigned right-shift operator

Method-level AsyncMethodBuilder

Native sized integers

Unmanaged generic structs

GRAMMES
25 20 15 10 5 0
PORTEE MINIMUM
PORTEE MAXIMUM 2 KILOG
LE POIDS LU SUR LE CADRAN NEST EXACT QUE SI L'AIGUILLE
SE TROUVE EN FACE DU ZÉRO LORSQUIL NY A AUCUN POIDS DANS LES PLATEAUX
AVERY
Societe Belge des Balances & Bascules
RUE DE L'INTENDANT, 43 - BRUXELLES
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